

Phenix PHIL Parameters and Atom Selections

1 documentation pages

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1. The Phenix atom selection syntax

The Phenix atom selection syntax

atom_selections.html

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Introduction

Many of the programs in Phenix, and phenix.refine in particular, allow (or require) you to specify selections of atoms in a model. Common examples include:

TLS (constrained anisotropic displacement) and rigid-body refinement groups
Selections for isotropic versus anisotropic B-factor refinement
Atoms to leave out for omit map calculation
Subset of atoms to modify properties for (in PDBTools)
Harmonically restrain atoms during refinement

- TLS (constrained anisotropic displacement) and rigid-body refinement groups
- Selections for isotropic versus anisotropic B-factor refinement
- Atoms to leave out for omit map calculation
- Subset of atoms to modify properties for (in PDBTools)
- Harmonically restrain atoms during refinement

All of these parameters are defined using a common syntax similar to what is used in CNS or PyMOL. The syntax allows for selection of atomic properties such as atom name, residue number, chain ID, or B-factor, which can be combined by the use of simple boolean statements (and, or, or not).

Video Tutorial

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- Basic overview
- Examples for selection syntax
- Examples for combined expressions
- How to test if an expression works (phenix.pdb_atom_selection)

Examples for selection expressions

All atoms

```
all
```

All C-alpha atoms (not case sensitive)

```
name ca
```

All atoms with H in the name (* is a wildcard character)

```
name *H*
```

Atoms names with * (backslash disables wildcard function)

```
name o2\*
```

All atoms in 'A' conformation

```
altloc A
```

Atom names with spaces

```
name 'O 1'
```

Atom names with primes don't necessarily have to be quoted

```
name o2'
```

Boolean "and", "or", and "not"

```
resname ALA and (name ca or name c or name n or name o)
chain a and not altid b
resid 120 and icode c and model 2
segid a and element c and charge 2+ and anisou
```

Note that the and, or, and not operators have equal priority, so parentheses may be required to indicate which clauses go together. The first example above selects for all atoms with the specified names in alanine residues, but if the parentheses are omitted:

```
resname ALA and name ca or name c or name n or name o
```

it will instead select C-alpha atoms in ALA, plus all atoms named C, N, or O regardless of residue name.

Also note that the outcome of boolean expression can be somewhat different from what one may expect comparing it with natural language. For example, if one wants to select ALA and GLY residues in the molecule, the selection should be

```
resname ALA or resname GLY
```

because suitable residues have residue name ALA or GLY. On the contrary, selection

```
resname ALA and resname GLY
```

will result in empty selection because there is no residue that can satisfy both conditions simultaneously.

A single residue by number

```
resseq 188
```

Note that if there are several chains containing residue number 188, all of them will be selected. To be more specific and select residue 188 in particular chain:

```
chain A and resid 188
```

this will select residue 188 only in chain A.

A range of residues

```
resseq 2:10
```

This selects all residue numbers falling within this range, independent of their order in the PDB file. The numbering must be ascending; resseq 10:2 will not work.

Insertion codes

resid combines the residue number (resseq) with the insertion code (icode); thus the following selections are identical:

```
resid 188
resseq 188 and icode ' '
```

as are these:

```
resid 27C
resseq 27 and icode 'C'
```

Specifying ordered ranges of residues

For some purposes, such as specifying TLS groups, it may be impractical to specify a numerical range of resseq attributes. This is especially the case when the residue numbering is not continuous and ascending, which is found in some PDB entries, or when the insertion code is non-blank. An alternative is to define a series of residues by their combined number and insertion code, using the through keyword:

```
chain A and resid 1000 through 17J
```

In this case, the residues will be examined in the order that they appear in the PDB file, and every atom falling between 1000 and 17J in chain A will be included. This is the only situation where the ordering of atoms is important in atom selections.

B-factors and occupancies

You may also select atoms based on their numerical properties:

```
bfactor > 80
bfactor = 0
occupancy < 1
occupancy = 0
```

Other

A few additional keywords are supported:

```
element H
water
pepnames
hetero
```

These examples specify hydrogen atoms, water molecules (detected by residue name), common amino acid residues, and atoms labeled as HETATM, respectively.

Selecting no atoms

```
(not all)
```

The default value of most selections is None (which in the GUI will appear as a blank field), but the treatment of this differs depending on how the atom selection will be used. For omit selections, None is equivalent to (not all); for rigid-body selections, phenix.refine will default to treating the entire model as a single rigid body. For B-factor refinement, leaving the selections set to None will result in their parameterization being determined based on the input model and atom type (hydrogens are treated specially).

Graphical selection

In the phenix.refine GUI, any valid atom selection can be visualized if you have a suitable graphics card and have already loaded a PDB file with valid symmetry information. The graphics window can be opened by clicking the "View/pick" button next to any atom selection field. Depending on the size of your structure, it may take several seconds for PHENIX to determine the atomic connectivity. The current selection, if any, will be highlighted:

The View/pick button opens a window that allows you to type in selections and immediately visualize the results, including the number of atoms selected. On mice with at least two buttons, clicking on an atom with the right button will open a menu for selecting residue ranges or chains. However, we recommend that you learn the selection syntax, as it is much more flexible than mouse controls. Selections made in the graphics window will be sent automatically to the appropriate control.

Once this window is open, it does not need to be closed; clicking a different "View/pick" button will transfer control of the display. (On Linux, you will first need to open the Actions menu and click Restore parent window to transfer mouse and keyboard controls back to the configuration dialog.)

"Smart" selections

In some contexts (primarily in phenix.refine and PDBTools), the geometry restraints information is used to extend the allowed keywords:

```
resname ALA and backbone
resname ALA and sidechain
peptide backbone
rna backbone or dna backbone
water or nucleotide
dna and not (phosphate or ribose)
within(5, (nucleotide or peptide) backbone)
```

Note however that these options cannot be visually selected in the Phenix GUI at present, and they may not be recognized by all programs.